

DEMOTHINE INJECTION

DESCRIPTION: DEMOTHINE INJECTION is a clear, colourless and preservative free liquid.

COMPOSITION:

DEMOTHINE INJECTION 100 MG/2ML: Each ampoule contains Pethidine Hydrochloride 100 mg per 2ml.

DEMOTHINE INJECTION 50 MG/ML: Each ampoule contains Pethidine Hydrochloride 50 mg per ml.

PHARMACODYNAMICS: Pethidine is a narcotic analgesic and sedative with actions quantitatively similar to those of morphine. After parenteral administration, 75 to 100 mg of Pethidine has comparable analgesic effects to 10 mg of morphine. Pethidine has a more rapid onset of action but shorter duration of analgesic effect than does morphine. Equiv-analgesic doses of Pethidine Hydrochloride and Morphine Sulphate produce the same degree of respiratory depression. However, the duration of this effect may be shorter with Pethidine. Its depressant effect of the cough reflex may be less than morphine at equiv-analgesic doses. Pethidine has spasmolytic as well as spasmogenic effects and may produce less constipation than morphine.

PHARMACOKINETICS: Pethidine undergoes rapid first-pass metabolism after oral administration and although well absorbed from the gastrointestinal tract, it is considerably less effective orally than parenterally. Peak analgesia occurs approximately 40 to 60 minutes after subcutaneous administration and 30 to 50 minutes after intramuscular injection. Analgesia may persist for 2 to 4 hours.

Pethidine is mainly metabolized in liver, primarily undergoing hydrolysis to pethidinic acid followed by partial conjugation with glucuronic acid. Other metabolic pathways include demethylation to norpethidine, which is hydrolysed to norpethidinic acid and then conjugated to form the glucuronide. Approximately 5% is excreted in the urine unchanged with acidification of the urine enhancing the excretion of unchanged pethidine.

Pethidine has an average plasma half-life of approximately 6.9 hours, a volume of distribution of 4 L/kg and is approximately 70% plasma protein bound.

INDICATION: Pethidine Hydrochloride injection is indicated for the relief of moderate to severe pain not responsive to non-narcotic analgesics. It may also be used as pre-operative medication and an analgesic adjunct in general anaesthesia and to provide obstetrical analgesia.

RECOMMENDED DOSAGE: Pethidine Hydrochloride may be administered by subcutaneous, intramuscular or slow intravenous injection. For intravenous administration, the dosage should be decreased and the injection administered very slowly as a dilute solution. When Pethidine is given parenterally, especially by the intravenous route, the patient should be lying down. When administered intravenously, a narcotic antagonist and facilities for assisted or controlled respiration should be immediately available.

Adults: Analgesia: 25 to 100 mg by s.c. or i.m. injection or 25 to 50 mg by slow i.v. injection every 3 to 4 hours.

Premedication: 50 to 100 mg by s.c. or i.m. injection or 25 to 50 mg by slow i.v. injection.

Obstetrics analgesia: 50 to 100 mg by s.c. or i.m. injection repeated 3 to 4 times at intervals of 1 to 3 hours if necessary.

Renal and hepatic dysfunction: Dosage reduction may be necessary in patients with renal or hepatic impairment.

Paediatric: Analgesia: 0.5 to 2 mg per kg bodyweight i.m. (maximum 100 mg) every 3 to 4 hours.

Premedication: 1 to 2 mg per kg bodyweight i.m. (maximum 100 mg)

ROUTE OF ADMINISTRATION: For s.c. / i.m. / slow i.v. injection

CONTRAINDICATIONS:

- Respiratory depression or where respiratory reserve is depleted.
- Head injury, raised intracranial pressure, brain tumour.
- Acute bronchial asthma, chronic airway disease.
- Cardiac arrhythmias, especially supraventricular tachycardias, cor pulmonale.
- Concurrent use of MAOIs, including selegiline or the use of MAOIs within 2 weeks prior.
- Convulsive states such as status epilepticus, tetanus and strychnine poisoning due to the stimulatory effects of pethidine on the spinal cord.
- Diabetic acidosis where there is a danger or coma.
- Acute alcoholism or delirium tremens.
- Severe liver disease, incipient hepatic encephalopathy.
- Hypersensitivity to pethidine.

PRECAUTIONS & WARNINGS:

Warnings. Narcotic analgesics have abuse potential. Psychological and physical dependence may occur with repeated dosing. Except in patients with terminal conditions, pethidine should be restricted to short-term administered for the relief of severe pain not responding to non-narcotic analgesics.

Abrupt withdrawal of pethidine in those physically dependent may precipitate withdrawal syndrome, including convulsions.

Use with extreme caution in patients with head injury and increased intracranial pressure. Respiratory depressant effects and ability to increase cerebrospinal fluid pressure may be exaggerated, and the clinical courses obscured.

Therapeutic doses of narcotics may decrease respiratory drive and increase airway resistance to the point of apnoea in patients with acute asthma, chronic obstructive pulmonary disease or cor pulmonale, or those with a substantially decreased pulmonary reserve or respiratory depression. Resuscitative equipment and narcotic antagonists must be readily available.

Precautions. Pethidine may increase ventricular response rate through a vagolytic action. The drug should therefore be used with caution in patients with atrial flutter and other supraventricular tachycardias. The administration of Pethidine may result in profound hypotension in patients whose difficulty to maintain blood pressure is compromised by depleted blood volume or concurrent administration of certain anaesthetics or phenothiazines. Pethidine may also produce orthostatic hypotension in ambulatory patients. Caution and an initial reduction in dose is recommended in patients who are elderly or debilitated and those with severely impaired pulmonary, hepatic or renal function and those with hypothyroidism, adrenocortical insufficiency, prostatic hypertrophy or urethral stricture. Pethidine may obscure the diagnosis and clinical course in patients with acute abdominal conditions, and should be used with caution in those with obstructive bowel disorders, or ulcerative colitis. Seizures may result from high doses. Patients with known seizure disorders should be carefully observed, especially when doses of pethidine are increased in response to tolerance.

Since Pethidine may cause drowsiness and general impairment of coordination, ambulatory patients should be cautioned against driving or operating machinery. If necessary, reduced doses of Pethidine Hydrochloride may be given intravenously. Such injections are to be administered slowly as diluted solutions. Rapid injection can increase the incidence of adverse reactions. Pethidine should not be administered intravenously unless a narcotic antagonist and the facilities for assisted or controlled respiratory are immediately available.

Risks from Concomitant use with Benzodiazepines

Profound sedation, respiratory depression coma, and death may result from the concomitant use of Pethidine with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine with an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Pethidine is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse or misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (*see Drug Interactions*).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Demothine Injection with serotonergic drugs (*See Interactions with Other Medicaments*).

This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (*See Interactions*).

with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Demothine Injection if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

DRUG INTERACTIONS: The depressant effects of pethidine are potentiated by other CNS depressants such as alcohol, sedatives, antihistamines, neuroleptics (e.g. phenothiazines, butyrophenones) tricyclic antidepressants and general anaesthetics. The actions of Pethidine on the gastrointestinal tract may influence absorption of other drugs.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids and benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Demothine Injection if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

PREGNANCY AND LACTATION:

Use in pregnancy. Narcotic analgesics may cause respiratory depression in the newborn infant. These products should only be used during labour after weighing the needs of the mother against the risk to the foetus. Animal reproduction studies have not been conducted with Pethidine Hydrochloride and safe use in pregnancy prior to labour has not been established in respect to possible adverse effects on foetal development. Infants born of mothers who have been taking pethidine chronically may exhibit withdrawal symptoms.

Use in lactation: Pethidine appears in breast milk. While the concentrations in breast milk following usual therapeutic doses in the mother may not be clinically significant, Pethidine administration to nursing mothers is not recommended.

SIDE EFFECTS: As with other narcotic analgesics respiratory depression is the major hazard of parenteral pethidine therapy. Other adverse experiences include:

More common: Light headedness, dizziness, sedation, nausea, vomiting, constipation and sweating. These effects seem to be more prominent in ambulatory patients and those not experiencing severe pain and may be relieved by reducing the dose slightly and lying the patient down.

Other: Central nervous system: Euphoria, dysphoria, weakness, headache, delirium, insomnia, anxiety, agitation, tremor, drowsiness, uncoordinated muscle movements, transient hallucinations, disorientation, visual disturbances, miosis.

Gastrointestinal: Dry mouth, anorexia, biliary tract spasm.

Cardiovascular: Facial flushing, tachycardia, bradycardia, palpitations, faintness, syncope, orthostatic hypotension.

Hypersensitivity: Pruritus, urticaria and other skin rashes, oedema.

Miscellaneous: Pain at injection site, local tissue irritation and induration following s.c. administration.

Postmarketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

SYMPTOMS AND TREATMENT OF OVERDOSE: **Symptoms:** Overdosage is characterized by respiratory depression with or without concomitant CNS depression. Also extreme somnolence progressing to stupor or coma, skeletal muscle flaccidity, cold and clammy skin and sometimes bradycardia and hypotension. In severe overdosage, apnoea, circulatory collapse, cardiac arrest and death may occur.

Treatment: Since respiratory arrest may result either through direct depression of the respiratory center or as the result of hypoxia, primary attention should be given to the establishment of adequate respiratory exchange through provision of a patent airway and institution of assisted or controlled ventilation. The narcotic antagonist, naloxone is a specific antidote. Naloxone should be administered intravenously, simultaneously with respiratory resuscitation. As the duration of effect of naloxone may be considerably shorter than that of pethidine repeated administration may be necessary.

Pethidine injection should not be self-administered and only given by physician.

Please tell your doctor, nurse or midwife if you are taking, or have recently taken, any other medicine including medicines obtained without prescription.

INCOMPATIBILITIES: Pethidine has been reported to be physically or chemically incompatible with solutions containing aminophylline, a barbiturate, heparin sodium, methicillin sodium, morphine sulphate, nitrofurantoin, phenytoin sodium, sodium bicarbonate, sodium iodide or sulphafurazole diethanolamine.

STORAGE CONDITIONS: Store below 30°C. Protect from light. KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.

SHELF LIFE: Please refer outer package.

PACK SIZES:

DEMOTHINE INJECTION 100 MG/2ML: Pack in 10 ampoules x 2 ml in a box,

DEMOTHINE INJECTION 50 MG/ML: Pack in 10 ampoules x 1 ml in a box,

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59, Kawasan 3, Taman Klang
Jaya, 41200 Klang, Selangor Darul Ehsan, Malaysia.